ALOK SINGH KATARIA

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GitHub: https://github.com/alokkataria101/quantum-physics

PHYSICS · COMPUTATIONAL SCIENCE · SOFTWARE ENGINEERING · AI · QUANTUM COMPUTING

OBJECTIVE

Seeking a role in **academic research, EdTech, or academic software development** where I can combine my foundation in **Physics, Mathematics, and Numerical Methods** with hands-on software engineering and DevOps experience. Specifically interested in opportunities involving **computational science, scientific software, educational platforms, or research tooling** — where rigorous mathematical thinking, numerical simulation, and modern software practices intersect to solve meaningful problems in academia or education.

PROFILE

Physicist and Senior Software Engineer with 25+ years of professional experience across software development, cloud platforms, DevOps, automation, enterprise systems, and EdTech. **IIT Kanpur Physics graduate** with a strong foundation in quantum mechanics, mathematical physics, linear algebra, probability, and numerical methods.

Hands-on experience with Python, scientific computing, AI/ML, and quantum-computing tools. Independent technical work includes **quantum mechanics, numerical methods, Hamiltonian and eigenvalue problems, VQE, quantum simulation, quantum chemistry, and quantum optimization** using Python and Qiskit.

Combines Physics and mathematical foundations with extensive software-engineering experience to **model, simulate, analyze, and prototype computational solutions**. Also experienced in developing mathematical and educational software and working with AI/ML approaches in EdTech.

Interested in opportunities involving Physics, computational science, scientific software, EdTech, AI/ML, and practical applications of quantum computing.

CORE AREAS

`Computational Physics · Quantum Mechanics · Mathematical Physics`
`Scientific Computing · Numerical Methods · Linear Algebra · Optimization`
`Quantum Computing · Quantum Algorithms · Quantum Simulation`
`VQE · Quantum Chemistry · Quantum Optimization`
`Quantum Noise & Error Mitigation`

SELECTED QUANTUM & COMPUTATIONAL PHYSICS WORK

- Applied classical numerical methods to quantum systems including the **hydrogen atom, 1-D particle-in-a-box, and harmonic oscillator**, using Python, NumPy, SciPy, linear algebra, and the **Finite Difference Method (FDM)** to construct Hamiltonians and compute energy eigenvalues, comparing numerical results with analytical solutions where available.
- For systems without convenient closed-form solutions, explored approximate numerical approaches using Python and **AI-assisted computational methods** for formulation, implementation, and validation.
- Applied a similar computational approach to quantum-computing formulations, using **qubit representations, Hamiltonians, quantum circuits, and VQE** with Python and Qiskit.
- Explored quantum-chemistry workflows using **STO-3G, Hartree–Fock, molecular Hamiltonians, and variational ansätze** for ground-state energy estimation, comparing results with classical calculations.
- Used classical numerical solutions as reference points for exploring corresponding **quantum formulations and variational approaches**, along with quantum optimization and quantum-information concepts such as portfolio optimization, qubit encoding, quantum randomness, spin dynamics, Hadamard transformations, and measurement probabilities.

PROFESSIONAL EXPERIENCE

IBM — Advisory Software Engineer | DevOps / SRE | 2022 – Present

1. CSUtils Ecosystem — Platform Architecture, Resilience & L3 Engineering

- Led architecture modernization and resilience engineering across IBM Cloud security utilities, modernizing Kubernetes configuration with CRDs, strengthening VPN connectivity with fail-fast validation, and ensuring reliable syslog delivery across pod restarts.
- Designed automated multi-cluster chaos testing and owned L3 production troubleshooting across Kubernetes, VPN, TCP/UDP connectivity, and networking failures to improve service reliability.

2. Dreadnought — IBM Cloud PaaS Re-platforming & Cloud Engineering

- Drove IaC-based provisioning and re-platforming of IBM Cloud infrastructure, including VPC networking, OpenShift, IAM, managed databases, and CI/CD integrations using standardized platform configurations.
- Implemented private endpoint connectivity, PAG/GPVPN integration, bastion access, and security scanning; supported complex cloud networking and platform escalations through root-cause analysis and remediation.

3. Compliance Automation — AI Agent, Auditree & CI/CD Engineering

- Led Auditree onboarding and Python-based compliance automation for SOC 2-aligned evidence collection, security posture assessment, centralized evidence storage in IBM Cloud

COS, and Git-based remediation workflows.

- Initiated an AI agent for troubleshooting Jenkins and Auditree pipelines, leveraging repository documentation for failure analysis and remediation recommendations; integrated automated workflows with CI/CD practices spanning Jenkins and Tekton.

Wipro — Lead Software Developer | 2022

Software development for the **CreditMemo** process of the Commissioning System for **Telefónica Germany**, building Python/Flask/Pandas services for data loading, aggregation, and validation.

- Implemented feature enhancements, defect fixes, and unit testing; worked with PostgreSQL and Google Cloud.

RevenueBox — Senior Software Engineer | 2021

B2B Account-Based Marketing (ABM) and go-to-market software focused on identifying, prioritizing, and engaging the right customer accounts.

- Developed a Python microservice to identify, process, and select relevant customer accounts based on multiple input signals.
- Designed a mathematical scoring model to calculate and assign a "**Quibble Score**" for account prioritization and tracking, using Python, Docker, MongoDB, and Google Cloud.

Lumos Learning — Academician / Educator & Research Scientist | 2019 – 2020

EdTech platform delivering dynamic, animated, and guided-practice learning experiences for **K-12 students**, aligned to Common Core Standards (corestandards.org), with emphasis on mathematics, automated question generation, and illustrative scientific concepts.

Drove the platform's modernization end to end — uniquely bringing **Physics, Mathematics, animation-driven software development, and serverless cloud architecture** together in one frame — and led the addition of several new modules spanning dynamic question creation, guided practice, and a mathematics chatbot.

- Led development of **Python-based animated and interactive question-generation modules** that expose K-12 students to dynamic, visually engaging problems with **varied difficulty levels** — catering to the full ability spectrum, from remedial to advanced.
- Architected and built **guided-practice workflows** with automated solution processing, difficulty estimation, and adaptive evaluation, enabling step-by-step problem solving aligned to Common Core standards.
- Led the design of a **mathematics-focused chatbot** and educational automation initiatives, applying NLTK-based natural language processing and classical neural-network/machine-learning techniques of the period.
- Designed the **serverless cloud architecture** using Python, AWS Lambda, and SQL Server to support these modules.
- Illustrated advanced mathematical and scientific concepts — including **solving quantum Hamiltonians via the Finite Difference Method and Fourier Transform techniques** — in simplified, visual, and accessible ways suitable for curious students and educators.

HCL Technologies — Senior Software Engineer | 2017 – 2018

L3 product support for **IBM AppScan** (web application security testing), troubleshooting complex customer issues, root-cause analysis, and coordinating fixes across SAST/DAST scanning and reporting workflows.

- Worked on AppScan SAST/DAST capabilities, customer configurations, defect analysis, and production issue resolution.

IBM India Software Labs — Senior Staff Software Engineer | 2008 – 2017

L3 Development & Troubleshooting Engineer for **IBM InfoSphere Data Replication (IIDR)** (previously DataMirror), specializing in complex production issues, root-cause analysis, debugging, and customer escalations.

- Expert in heterogeneous data replication and integration, including **DB2 for i / AS400, Oracle, DB2, SQL Server, and Teradata**, with IBM messaging and remote journaling technologies.
- Strong experience in cross-language and cross-platform integration, including **AS400/IBM i** and COBOL-based enterprise systems; supported customer ETL and data-movement workflows.
- Recognized as a performance specialist, contributing to performance analysis and optimization and receiving an **IBM award for performance expertise**.
- Contributed to feature development, product maintenance, automation, and build/release pipeline engineering across multiple product releases.
- Worked with C, Java, Python, RPG, COBOL, AS400, and Jenkins.

CA Technologies — Software Engineer | 2005 – 2008

Worked on **eTrust Access Control**, **eTrust Audit**, and **eTrust Single Sign-On** — enterprise security products.

- Developed more than 100 automated tests for nightly builds and ported product code to AIX.
- Investigated customer and QA defects including memory corruption, enhancements, and installation/upgrade issues.
- Worked with C, Python, Linux, UNIX, AIX, Solaris, and HP-UX.

TECHNICAL SKILLS

Programming: Python, C, Java, Shell, JavaScript

Scientific Computing: NumPy, SciPy, Pandas, MATLAB, scikit-learn, TensorFlow

Quantum Computing: Qiskit, VQE, Quantum Circuits, Quantum Simulation

Cloud & DevOps: IBM Cloud, AWS, Google Cloud, Kubernetes, OpenShift, Docker, Jenkins, Tekton, CI/CD

Databases: PostgreSQL, MongoDB, SQL Server, Oracle, DB2, Teradata

Operating Systems: Linux, AIX, Solaris, HP-UX, Windows, IBM i / AS400

SELECTED RESEARCH & TECHNICAL WORK

Quantum Intuition: From Spin Dynamics to the Hadamard Gate and π Estimation via Quantum Randomness

Explored quantum randomness and quantum-state transformations through Qiskit-based simulations, including spin dynamics, Hadamard transformations, measurement probabilities, and Monte Carlo estimation of π.

Status: Under IBM internal review for publication.

Classical VQE Demonstration for the Hydrogen Atom

Developed a classical demonstration of the VQE approach for the hydrogen atom, connecting the variational principle, Hamiltonian formulation, parameterized states, energy expectation values, and numerical optimization.

Status: Under IBM internal review for publication.

INTELLECTUAL PROPERTY

- **US Patent #9,528,562** — Granted US patent related to mechanics.
- Two additional mechanics-related inventions under IBM patent review.

EDUCATION

IIT Kanpur — Integrated M.Sc. in Physics | 2000

Academic foundation includes Quantum Mechanics, Classical Mechanics, Electromagnetism, Mathematical Physics, Linear Algebra, Probability, Differential Equations, Numerical Methods, and Statistical Physics.

ACADEMIC DISTINCTIONS & AWARDS

- **All India IIT-JEE Advanced Rank** — 1211 (General)
- **GATE All India Rank** — 29 (General) | **Score** — 98.73 (General)
- **Roorkee Entrance Exam 1994** — Rank 1741 (General)
- **Bihar Engineering Entrance Test** — Rank 452 (General)
- **IBM Excellence and Eminence Award (2012)** — Received for outstanding L3 support, troubleshooting expertise, and high-priority resolution of complex client issues.

PHYSICS TEACHING & MENTORING | IIT-JEE ADVANCED / NEET FACULTY

Institutions: Vizag Vikas College · Narayana Junior College, Kurnool · SRM · PIE Education

- **Vizag Vikas College (2001–2004) — IIT-JEE Results & Academic Excellence:** Mentored students at a relatively small college, contributing to 40+ IIT-JEE selections, including an All India Rank (AIR) of 155, with student ranks extending to approximately AIR 2,000. Also contributed to an EAMCET rank of 20. Designed original, challenging IIT-JEE Advanced-level questions and supported competitive examinations and faculty recruitment across Andhra Pradesh.
- **Narayana Junior College, Rajiv Vihar, Kurnool (2017–2018) — Academic Results & Question-Quality Improvement:** Contributed to the branch's early IIT-JEE selection results, including approximately four General-category selections and additional reserved-category selections, alongside AIEEE and EAMCET successes. Reviewed and refined questions derived from Narayana's curriculum and published textbooks, identifying errors and improving accuracy, originality, clarity, and difficulty calibration.
- **NEET & CBSE Physics — SRM:** Taught and mentored students in Physics for medical entrance examinations and CBSE curricula, emphasizing conceptual understanding and systematic problem-solving.
- **Teaching Methodology:** Built students' understanding from fundamental Physics and mathematical principles to advanced IIT-JEE-level applications, emphasizing first-principles derivations, mathematical formulation, conceptual depth, and rigorous problem-solving through classroom demonstrations.
- **Early Recognition:** Received an award from Brilliant Tutorials during Class XII in recognition of problem-solving ability.

RESEARCH & CODE | COMPUTATIONAL AND QUANTUM PHYSICS

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Independent computational physics projects exploring quantum mechanics through Python, numerical methods, and Qiskit. Selected work includes numerical solutions of the Schrödinger equation, Hamiltonian modelling and eigenvalue analysis, quantum circuit simulation, the Variational Quantum Eigensolver (VQE), quantum chemistry, quantum optimization, and quantum-information experiments.